

Gandharva Appagere

EXPERIMENTAL PHYSICIST • DETECTORS, DATA & AI

PhD candidate in experimental nuclear & hadron physics, Stockholm University

Stockholm

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PROFILE

PhD candidate in experimental nuclear and hadron physics at Stockholm University, studying hyperon–hyperon ($\Lambda\Lambda$) interactions with the HADES experiment at GSI and simulation-based feasibility studies for the CBM experiment within the QCD at FAIR programme.

EDUCATION

- 2021–2027
(expected) **PhD, Experimental Nuclear & Hadron Physics**
Stockholm University, Department of Physics (Fysikum) Stockholm, Sweden
Thesis: “Hyperon–hyperon interactions from proton–proton collisions with HADES and CBM”
Supervisors: Prof. Per-Erik Tegnér (main supervisor since Dec 2024); Dr. María Doncel Monasterio (main supervisor 2021–2024, now co-supervisor); Dr. Jenny Taylor, GSI (co-supervisor since 2025); Dr. Michael Papenbrock, Uppsala University (co-supervisor 2022–2025). Collaborations HADES and CBM at GSI/FAIR (Darmstadt, Germany); QCD at FAIR programme.
- 2021–2024 **Licentiate of Philosophy (Fil.lic.), Physics**
Stockholm University Stockholm, Sweden
Thesis: “ Λ - Λ hyperons: the road to estimate the production cross-section in proton-proton collisions”
Licentiate seminar 27 March 2024.
- 2018–2020 **MSc, Electromagnetics, Fusion and Space Engineering**
KTH Royal Institute of Technology Stockholm, Sweden
Thesis: “Can FIMPs be Dark Matter Candidates?”
Thesis advisor: Prof. Robert Mann (Perimeter Institute / University of Waterloo). Thesis on symmetry breaking with the Coleman–Weinberg effective potential at finite temperature.
- 2013–2017 **BE, Electronics and Communication Engineering**
BMS College of Engineering Bengaluru, India
Thesis: “CLAIRE: Cognitive Linguistic Analyser and Interpreter with Real-time Efficiency”

RESEARCH EXPERIENCE

- Aug 2021 – present **Doctoral researcher (PhD candidate)**
Stockholm University · HADES & CBM collaborations, GSI/FAIR Stockholm, Sweden & Darmstadt, Germany
Experimental hadron physics on the HADES spectrometer at GSI and simulation studies for the CBM experiment at FAIR.
- Main analysis responsibility in the HADES collaboration for the $\Lambda\Lambda$ channel in proton–proton collisions ($\sqrt{s} = 3.46$ GeV): event selection, reconstruction, simulation, signal extraction and systematic uncertainties.
 - Took detector shifts during the five-week February 2022 proton beam time and a 2025 pion-beam run; worked with drift-chamber online quality assurance.
 - Developed a relative time-of-flight particle-identification method that compensates start-detector timing problems, used with kinematic fitting and 2D template likelihood fits.
 - Trained multiclass machine-learning classifiers (ROOT TMVA) to separate $\Lambda\Lambda$ signal from background.
 - Built a staged (S0–S3) analysis pipeline with automated systematic variations on the GSI batch farm (Slurm), developed with AI coding assistants (Cursor, LLMs).
 - Wrote an event generator and detector-response folding for CBM feasibility studies of $\Lambda\Lambda$ cusps; contributed to the QCD at FAIR white paper.
 - Hosted and minuted the weekly meetings of the HADES Λ -Reconstruction Task Force and Kinematic-Fitting working group (2022–23).

C++ • ROOT • Python • TMVA • Slurm • PLUTO • GEANT • KinFit • Git

Jun-Aug 2019	Summer research project — density-matrix dynamics <i>KTH Royal Institute of Technology</i> Literature study on the dynamics of the density matrix and quantum entanglement. Supervision: Prof. Erik Aurell	Stockholm, Sweden
Oct 2018 – 2019	Research project — flux transfer events <i>KTH Royal Institute of Technology, Space and Plasma Physics</i> Statistical study of flux transfer events near Earth's magnetosphere using ESA Cluster spacecraft data, supported by NASA MMS data. Supervision: Asst. Prof. Anita Kullen, Prof. Tomas Karlsson	Stockholm, Sweden

INDUSTRY & ENGINEERING EXPERIENCE

Sep 2018 – 2019	Electric Propulsion Engineer <i>KTH Hyperloop team</i> Linear induction motor for the SpaceX Hyperloop Pod Competition 2019. • Designed a double-sided linear induction motor for the pod's propulsion; electromagnetic finite-element simulations in COMSOL and ANSYS Maxwell, with MATLAB. COMSOL • ANSYS Maxwell • MATLAB	Stockholm, Sweden
Aug 2017 – Jul 2018	Associate Software Engineer <i>Robert Bosch Engineering and Business Solutions (RBEI)</i> • Worked on signal-processing algorithms for automotive radar sensors used in driver-assistance systems, in an industrial software-engineering environment.	Bengaluru, India
Jan-Jul 2017	Intern, Network Engineering Facility <i>AT&T</i> VPN order-management processes in a network-engineering organisation.	Bengaluru, India
Aug 2016 – Jun 2017	Hybrid levitation for Hyperloop — engineering project <i>BMS College of Engineering</i> Design of air-bearing levitation combined with linear-induction-motor propulsion for a Hyperloop concept.	Bengaluru, India
Jul 2015	Intern, Quality Assurance division <i>ISRO — Laboratory for Electro-Optics Systems (LEOS)</i> Circuit analysis of star, Earth and Sun sensors for the ASTROSAT and Chandrayaan-2 missions.	Bengaluru, India

PUBLICATIONS

ARTICLES & PROCEEDINGS

- [1] **G. Appagere** (for the HADES and CBM Collaborations), “Investigating Double-Strange Systems at FAIR: $\Lambda\Lambda$ interactions and neutron star cores”, *JPS Conference Proceedings — Proceedings of the 15th International Conference on Hypernuclear and Strange Particle Physics (HYP2025), Tokyo (2026)*. **IN PRESS**
- [2] M. Frank, R. Mann, P. Poullose, **G. Appagere**, “Radiatively generated dark matter and Higgs scalar masses in a minimal extension of the Standard Model” (2026). **IN PREPARATION**

COLLABORATION PAPERS & REPORTS

- [3] J. G. Messchendorp, F. Nerling, ..., **G. Appagere**, et al., “Hadron Physics Opportunities at FAIR”, *arXiv:2512.15986 [hep-ex]* (2025). **PREPRINT**
Co-author; contributed the $\Lambda\Lambda$ cusp feasibility study for CBM to the baryon-baryon interactions chapter.

THESES

- [4] **G. Appagere**, “ Λ - Λ hyperons: the road to estimate the production cross-section in proton-proton collisions”, *Licentiate thesis, Department of Physics, Stockholm University* (2024).
- [5] **G. Appagere**, “Can FIMPs be Dark Matter Candidates?”, *MSc thesis, KTH Royal Institute of Technology (advisor: Prof. R. Mann)* (2020).

CONFERENCE TALKS

Sep 2026	“Cusp effects and near-threshold structures: $\Lambda\Lambda$ interaction simulation study with CBM” <i>NSTAR 2026 — 15th International Workshop on the Physics of Excited Nucleons, Seville, Spain</i>
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May 2026	“Hyperon–hyperon interaction studies close to threshold with CBM at FAIR — a feasibility study with the $\Lambda\Lambda$ production channel” <i>15th Nordic Meeting on Nuclear Physics</i> , Visby, Sweden
Mar 2026	“Probing hyperon–hyperon interactions with CBM at FAIR” <i>DPG Spring Meeting 2026, Hadrons and Nuclei section (HK 37.2)</i> , Erlangen, Germany CONTRIBUTED
Oct 2025	“Investigating double-hyperon systems using HADES and CBM: insights into Λ - Λ interactions and neutron star cores” <i>Annual Swedish Nuclear Physics and SFAIR Meeting 2025</i> , Chalmers, Göteborg, Sweden CONTRIBUTED
Oct 2025	“Investigating double-strange systems at FAIR: Λ - Λ interactions and neutron star cores” <i>HYP2025 — 15th International Conference on Hypernuclear and Strange Particle Physics</i> , Tokyo, Japan CONTRIBUTED
Oct 2024	“ Λ - Λ hyperons: journey towards the understanding of the hyperon–hyperon interactions in neutron stars” <i>Annual Swedish Nuclear Physicists’ and SFAIR Meeting 2024</i> , Lund, Sweden CONTRIBUTED
Oct 2023	“Investigating hyperon–hyperon interaction with the HADES experiment” <i>Annual Swedish Nuclear Physicists’ and SFAIR Meeting 2023</i> , Sweden CONTRIBUTED
Jun 2023	“Understanding the Λ - Λ interaction with the HADES experiment” <i>Fysikdagarna 2023</i> , Stockholm, Sweden CONTRIBUTED
Oct 2022	“ Λ - Λ study using a kinematic fitter in pp reactions at HADES” <i>Annual Swedish Nuclear Physicists’ and SFAIR Meeting 2022</i> , Sweden CONTRIBUTED

COLLABORATION PRESENTATIONS

Aug 2026	Extracting scattering parameters from $\Lambda\Lambda$ cusps in pp collisions at $\sqrt{s} = 3.59$ GeV · <i>CBM Exclusive Channels meeting</i> , online
Jun 2026	$\Lambda\Lambda$ reconstruction from pp at $\sqrt{s} = 3.46$ GeV — update on cross-section estimates and systematics · <i>HADES Analysis Meeting VIII</i> , Darmstadt, Germany
May 2026	Extracting scattering parameters from Λ - Λ cusps in pp collisions at $\sqrt{s} = 3.59$ GeV · <i>CBM Exclusive Channels meeting</i> , online
Feb 2026	Searching for double-strangeness production in pp collisions at $\sqrt{s} = 3.46$ GeV — update on Λ - Λ reconstruction · <i>50th HADES Collaboration Meeting</i> , GSI, Darmstadt, Germany
Feb 2026	Extracting scattering parameters from Λ - Λ cusps in pp collisions at $\sqrt{s} = 3.59$ GeV · <i>CBM Exclusive Channels meeting</i> , online
Nov 2025	Searching for double-strangeness production in pp collisions at $\sqrt{s} = 3.46$ GeV · <i>HADES Physics Working Group — Exclusive Channels</i> , online
Oct 2025	From HADES to CBM: investigating Λ - Λ interactions · <i>CBM Collaboration Meeting</i> , online
Jul 2025	$\Lambda\Lambda$ analysis — cusp analysis update · <i>CBM Exclusive Channels working group</i> , online
Jun 2025	$\Sigma^+\Sigma^+$ analysis — report from the GSI visit · <i>CBM Exclusive Channels working group</i> , online
Jun 2025	$\Sigma^+\Sigma^+$ analysis — progress report · <i>CBM Exclusive Channels working group</i> , online
Apr 2024	Status report on exclusive Λ - Λ reconstruction in pp at 4.5 GeV — towards a cross-section determination · <i>46th HADES Collaboration Meeting</i> , GSI, Darmstadt, Germany
Apr 2024	Status report on exclusive Λ - Λ reconstruction in pp at 4.5 GeV · <i>HADES Physics Working Group — Exclusive Channels</i> , online
Oct 2023	Status report on exclusive Λ - Λ reconstruction in pp at 4.5 GeV — reconstruction efficiencies compared with simulations · <i>HADES Collaboration Meeting</i> , Stockholm, Sweden
Jul 2023	Update on Λ - Λ reconstruction with minimum-bias analysis · <i>HADES Analysis Meeting</i> , Orsay, France
Jun 2023	Update on Λ - Λ reconstruction with minimum-bias analysis · <i>HADES Physics Working Group — Exclusive Channels</i> , online
Mar 2023	First Λ - Λ signals from $p(4.5\text{ GeV}) + p$ · <i>HADES Collaboration Meeting</i> , GSI, Darmstadt, Germany
Mar 2023	Inclusive- Λ reconstruction with KinFit and forward particle species — PID benchmark for forward tracks · <i>HADES Physics Working Group — Exclusive Channels</i> , online
Feb 2023	Analysis of reconstructed forward momentum · <i>HADES Common Λ-Reconstruction Task Force</i> , online
Dec 2022	Λ simulations and momentum fit for forward-momentum reconstruction · <i>HADES Physics Working Group — Exclusive Channels (Hyperons)</i> , online
Sep 2022	Λ - Λ study with a kinematic fitter in pp at 4.5 GeV: status report · <i>43rd HADES Collaboration Meeting</i> , Coimbra, Portugal

SCHOOLS, WORKSHOPS & EXPERIMENTS

EXPERIMENTS & BEAM TIMES

- 2025 HADES pion-beam run • GSI, Darmstadt, Germany
- Feb 2023 Nuclear-structure experiment JR160 at the JYFL Accelerator Laboratory (neutron-deficient Os isotopes) • Jyväskylä, Finland
- Feb 2022 HADES proton beam time (pp at 4.5 GeV) — detector shifts • GSI, Darmstadt, Germany

WORKSHOPS & RESEARCH VISITS

- Feb 2026 Research visit — Forschungszentrum Jülich (theory) and GSI • Jülich & Darmstadt, Germany
- Oct 2025 HAMP2025 — Hadron Physics using Antimatter Probes • KEK Tokai, Japan
- Jun 2025 QCD at FAIR workshop — contributions to the FAIR hadron-physics white paper • Catania, Italy
- 2024-2025 Research visits to the GSI hyperon group (CBM feasibility studies) • Darmstadt, Germany

SCHOOLS & COURSES

- Sep 2023 HaSP / STRONG-2020 Hadron Spectroscopy Summer School • Salamanca, Spain
- Oct 2022 Basic course in radiation safety, Stockholm University • Stockholm, Sweden

TEACHING & SUPERVISION

COURSES

- 2021-2025 **Nuclear Physics (MSc)**, FK7051 • HT2021, HT2023, HT2025
Laboratory supervisor — lab introduction lecture, lab compendium, grading of lab reports and oral presentations
- 2021-2023 **Programmering, numeriska metoder och statistik för fysiker**, FK4026 • HT2021, HT2022, HT2023
Laboratory assistant — Python labs, numerical-methods part, project grading
- 2024 **Detector Physics (MSc)**, FK7056 • HT2024
Laboratory supervisor — muon lifetime and HPGe gamma-spectroscopy labs
- 2024-2025 **Project in Experimental Modern Physics**, FK5026 • HT2024, HT2025
Laboratory supervisor — Compton-scattering lab
- 2025 **Elektromagnetism och vågor**, FK5019 • VT2025
Problem-solving sessions
- 2025 **Vågrörelselära och optik (Waves and Optics)** • HT2025
Exercise-group leader and grading of hand-ins

SUPERVISION

- 2022 **Bachelor's thesis in Medical Radiation Physics** • VT2022
Co-supervisor — Monte Carlo study of a high-resolution germanium (HPGe) detector at the AlbaNova low-background laboratory

PEDAGOGICAL TRAINING

- 2021 **Introduction to teaching (3 hp)**, Stockholm University
Higher-education teacher training

SERVICE & LEADERSHIP

- 2022-2026 **Chair** (2025–2026), **Vice-Chair** (2023–2025), **Board member** (2022–2023), Fysikum PhD Council, Stockholm University. Lead the department's doctoral-student council: run board and general meetings and represent PhD students in departmental bodies. Contributed to clearer departmental guidelines for changing supervisors and to an improved half-time review process; started PhD-student meetings in the Instrumentation division.
- 2024-2025 **Vice-Chair**, Central Doctoral Student Council (CDR), Stockholm University. University-wide representation of doctoral students; represented CDR in national doctoral-student bodies.
- 2024 **Vice-Chair**, Natural Sciences Faculty Doctoral Student Council (NFR), Stockholm University. Faculty-level doctoral-student representation; frequently meeting secretary.

2023–2026	Doctoral-student representative , Fysikum Department Board and committees (Research Education, Technical Division, Computer, Work Environment). Department Board member and committee seats on research education, technical-division and computing committees.
2025	Co-organiser , NFR Debate Forum, Stockholm University. Oxford-style debates for doctoral students across the Faculty of Science.
2021–2025	Lab responsible , Nuclear physics student laboratory, Fysikum. Responsible for HPGe detectors, scintillators and the radioactive-source inventory; lab rules and safety, a source booking and record system, and procurement of a proportional counter.
2024–2025	Mentee , Industry mentor programme for doctoral students. Paired with a senior industry mentor to prepare the transition from academia to industry.
2015	Executive Committee Secretary , IEEE Student Chapter, BMS College of Engineering.

GRANTS & AWARDS

GRANTS & SCHOLARSHIPS

2026	C F Liljevalch J:ors travel grant , Stockholm University. Conference travel.
2025	Donation scholarship, Faculty of Science , Stockholm University. Travel to HYP2025, Tokyo.
2022	Birger och Gurli Grundströms forskarstipendiefond i fysik — travel stipend , Department of Physics, Stockholm University. Participation in an experiment at the JYFL Accelerator Laboratory, Jyväskylä.
2022	Donation scholarship, Faculty of Science (August Emil Wilhelm Smitts stipendie- och understödsstiftelse) , Stockholm University. Travel to HADES collaboration meetings.

AWARDS

2017	SAE E-Baja 2017 — 3rd place in India, design review , SAE India.
2016	IEEE Darrel Chong Student Activity Award , IEEE. Special-interest-group activities teaching off-syllabus topics.
2015	SAE Aero Design West 2015 — 13th worldwide (team captain) , SAE International, Van Nuys, CA.

SKILLS

Data analysis, ML & AI	Statistical inference (maximum-likelihood & template fits, least squares, sideband methods) • Uncertainty quantification (systematic variations, correlated averages) • Kinematic (constrained) fitting (vertex and multi-constraint fits) • Machine-learning classification (multiclass classifiers in ROOT TMVA) • AI-assisted development (Cursor and LLM coding assistants for building and automating pipelines) • Machine Learning for Physicists and Astronomers (PhD course, 7.5 hp)
Programming & software	Python (NumPy, SciPy, pandas, matplotlib, uproot, iminuit, Jupyter) • C++ & ROOT (analysis frameworks, PyROOT, RooFit) • C • Git / GitLab • Bash & batch scripting • JavaScript, HTML, CSS • LaTeX
Simulation & modelling	Monte Carlo event generation (PLUTO, UrQMD, own generator with final-state interactions) • Detector simulation (GEANT3-based HGeant, CbmRoot) • Finite-element electromagnetics (COMSOL, ANSYS Maxwell) • Mathematica, MATLAB
Detectors & instrumentation	Tracking & timing detectors (drift chambers, straw tubes, RPC, LGAD start detector, time of flight) • Gamma spectroscopy (HPGe detectors, scintillators, neutron activation) • Beam-time operations & online data-quality monitoring • Radiation safety & radioactive-source handling • Automotive radar signal processing
Computing & HPC	HPC batch computing (Slurm on the GSI Virgo cluster) • Containers & software distribution (Singularity/Apptainer, CVMFS) • Large datasets (~155 TB raw data, Lustre, ROOT trees, HDF5)

LANGUAGES

English	Fluent — professional working proficiency
Swedish	Basic (A2, studying)